18. Certified optimal-transport bounds for strictly correlated electrons

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Electrons repel one another, but a prescribed electron density also constrains where they spend their time. Imagine arranging their joint positions so they avoid one another as effectively as possible while preserving that average density. This problem makes that arrangement question precise and seeks reliable upper and lower bounds on its repulsion energy. It represents a limiting strong-correlation model, not the full energy of ordinary electrons.

Key terms

- **Electron density** ρ — The average spatial distribution of electrons, integrating to their total number N.
- **Coulomb cost** — The repulsion contribution between charged particles, proportional here to the inverse of their separation.
- **Optimal transport** — Optimization over ways of distributing or coupling quantities while satisfying prescribed distribution constraints.
- **Marginal distribution** — The distribution of one variable after the other variables in a joint distribution are ignored.
- **Coupling** Γ — A joint distribution of all electron positions consistent with the prescribed individual marginals.
- **Multi-marginal transport** — Transport optimization with more than two prescribed marginals.
- **Strictly correlated electrons (SCE)** — A limiting model in which electron positions are organized to minimize interaction energy at fixed density.
- **Dual potential** — An auxiliary function providing a certified lower bound when it satisfies the required inequality everywhere.
- **Discretization error** — The difference introduced by replacing a continuous spatial problem with a finite representation.

Why this matters

Strongly correlated electrons are difficult to approximate because their motions are strongly linked. This limiting problem gives a mathematically structured reference for studying that linkage and for developing density-functional ideas. Verified bounds help distinguish a true property of the continuum model from an artifact of a spatial grid or optimization tolerance.

What progress would mean

A useful result would enclose the SCE interaction energy for a declared density. It could inform further electronic-structure theory, while leaving kinetic energy, finite-correlation effects and actual molecular reaction paths to additional modeling.

Core question

For a supplied normalized electron density, can a discretized Coulomb multi-marginal transport problem yield verified bounds, with continuum error separated from optimization error?

Scope and assumptions

Take an electron density $\rho \geq 0$ with $\int \rho = N$ and marginal $\mu = \rho/N$. Begin with N=2 and a compactly supported mathematical density or a finite rational discrete model. Specify treatment of the diagonal Coulomb singularity. A continuous-density result additionally needs domain-tail and discretization estimates.

The main target is the strictly correlated interaction functional. Wasserstein comparison of densities is a separate optional metric task; it is not a minimum-energy chemical reaction path.

Mathematical target

Define

$$V_{SCE}[\rho] = \inf_{\Gamma \in \Pi(\mu,\ldots,\mu)} \int \sum_{i<j} |r_i - r_j|^{-1} \, d\Gamma.$$

Γ is a probability measure and may be symmetrized; it is not an antisymmetric probability distribution. The symmetric dual uses u with $\Sigma_i\ u(r_i) \leq \Sigma_\{i<j\} |r_i - r_j|^{-1}$ and objective $\int u\rho$. A feasible coupling gives an upper bound and a dual-feasible potential gives a lower bound for the same problem. Define $W_\infty = V_SCE - U_H$ if subtracting Hartree energy; do not call V_SCE exact exchange.

Required outputs and proof obligations

Return marginal conventions, cost and singularity policy, primal plan, dual potentials, rigorously checked constraints, and $L \leq V \leq U$. State whether V is a finite discrete optimum or the continuum functional. Sinkhorn candidates must be converted to feasible bounds for the unregularized problem, with entropic bias controlled.

Validation and rejection controls

Verify a small rational-cost transport LP exactly and a one-dimensional quadratic-cost quantile example as a separate solver test. For Coulomb densities, use a case with a known co-motion construction under its actual hypotheses. Reject comparing normalized densities with unequal total mass unless normalization or an unbalanced-OT model is explicitly chosen.

Milestones and research extension

Baseline: certified two-marginal discrete bounds. Strong: one continuum enclosure with singularity/tail control. Research extension: N>2 with symmetry reduction and validated bounds, or a demonstrably improved relaxation. Do not promise deterministic Monge maps for arbitrary multi-marginal inputs.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Cotar, Friesecke and Klüppelberg, Density functional theory and optimal transportation with Coulomb cost — Strong-correlation/semiclassical connection; theorem hypotheses must be checked for each density.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\int \rho = N, \qquad \mu = \rho/N, \qquad c(r_1, \dots, r_N) = \sum_{i<j} |r_i - r_j|^{-1},$$

$$V_{\mathrm{SCE}}[\rho] = \inf_{\Gamma \in \Pi(\mu,\dots,\mu)} \int c \, d\Gamma, \qquad \sum_i u(r_i) \le c(r_1, \dots, r_N),$$

$$\underbrace{\int u\rho}_{L} \;\le\; V_{\mathrm{SCE}}[\rho] \;\le\; \underbrace{\int c \, d\Gamma_{\mathrm{feasible}}}_{U}.$$

The coupling is a probability measure; it may be symmetrized and is not antisymmetric. This is a Coulomb interaction functional, not exact exchange. For a discrete problem, the same inequalities apply only to the discrete optimum until continuum error is bounded.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. “Blocking” means the original issue can invalidate the scientific target or claimed certificate;

"major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — wrong functional (16, 34, 98–108).** Coulomb OT describes a strong-correlation limit, not the exact exchange functional asserted here. The negative half-cost formula and negative-cost dual are inappropriate for the positive repulsion minimization.
2. **Major — normalization and antisymmetry.** The electron density integrates to N, while each probability marginal integrates to one. Fermionic wavefunction antisymmetry does not make a probability measure antisymmetric.
3. **Major — reaction pathway inference (18, 36).** A Wasserstein geodesic minimizes transport cost, not a molecular potential-energy barrier. The original chemical interpretation lacks a dynamical/energetic bridge.
4. **Major — uniqueness and certification (106, 868–905).** Quadratic-cost Brenier results do not transfer automatically to Coulomb multimarginal problems. Marginal residuals and a regularized solver output are not continuum certificates. The revised formulation distinguishes these errors and removes the inappropriate helium-exchange target.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a supplied normalized electron density, can a discretized Coulomb multi-marginal transport problem yield verified bounds, with continuum error separated from optimization error?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: certified two-marginal discrete bounds. Strong: one continuum enclosure with singularity/tail control. Research extension: $N>2$ with symmetry reduction and validated bounds, or a demonstrably improved relaxation. Do not promise deterministic Monge maps for arbitrary multi-marginal inputs.

Sources supporting the corrections

- Cotar, Friesecke and Klüppelberg, Density functional theory and optimal transportation with Coulomb cost — Strong-correlation/semiclassical connection; theorem hypotheses must be checked for each density.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise “publication-quality” in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.